Salts and constituents

Overview

SWAT+ tracks two optional families of conservative-but-reactive constituents:

- Salt ions (`rtb salt`): SO4, Ca, Mg, Na, K, Cl, CO3, HCO3 (eight ions)
- Trace constituents (`rtb cs`): SeO4, SeO3, boron

Both families share the same flow paths as nitrate: surface runoff, lateral flow, tile drainage, percolation, plant uptake, rainfall and dry deposition, irrigation, and fertiliser/amendment inputs. Salt ions are subject to equilibrium chemistry (precipitation-dissolution of CaCO3, MgCO3, CaSO4, MgSO4, NaCl and cation exchange between Na, Ca, Mg, K). Selenium and boron use simple linear sorption and a first-order reaction term. Salinity stress can reduce crop water uptake.

These modules are active only when their respective databases are present, controlled by `cs_db%num_salts` and `cs_db%num_cs` (set during `cs_read_*` setup).

Process equations

Soil-water transport

For both salts and trace constituents the layer-level transport routine mirrors `nut_nlch.f90`. For each salt ion (or cs species) `m`:

```
vv = perc + flat + (surfq if layer 1) + (qtile if tile layer)
co = mass * (1 - exp(-vv / ul)) / vv
surq_mass  = co * surfq             # layer 1 only
latq_mass  = co * flat
tile_mass  = co * qtile             # at tile layer
perc_mass  = co * prk               # next layer or out of profile
```

Lateral, tile, and percolation are first-order in mobile water with porosity `ul`. No partition coefficient is applied to salt ions; sorbed selenium and boron use `cs_sorb_hru.f90`. Source: `salt_lch.f90`, `cs_lch.f90`.

Salt chemistry in soil

`salt_chem_hru.f90` and `salt_chem_soil_single.f90` solve equilibrium speciation per layer using:

- Activity coefficients via ionic strength (Davies / extended Debye-Huckel form)
- Precipitation-dissolution of five minerals: CaCO3 (Ksp = 3.07e-9), MgCO3 (4.79e-6), CaSO4 (7.89e-5), MgSO4 (7.24e-3), NaCl (37.3)
- Complexation reactions (CaCO3, MgCO3, CaSO4, MgSO4, NaCl)
- Cation exchange between Na, Ca, Mg, K through `cationexchange`

These run iteratively until ion concentrations converge. The same scheme is applied to the aquifer in `salt_chem_aqu.f90` using a separate set of Ksp values (currently the same defaults).

Salinity stress on plants

When `salt_tol_sim = 1` and at least one salt ion is simulated, `pl_waterup.f90` computes a salinity reduction factor per layer using the FAO yield-vs-ECe relationship:

```
soil_TDS_sat = sum(soil_salt_conc) at saturation
soil_ECe = soil_TDS_sat / salt_tds_ec
if (soil_ECe > a):
    reduc_salt = (100 - b * (soil_ECe - a)) / 100
```

a is the salt tolerance threshold (dS/m), b is the slope (yield loss per dS/m above the threshold). Both are per-plant parameters read from salt_plants.slt. If salt_soil_type = 1 (CaSO4 / gypsiferous), the threshold is increased by 2 dS/m. The minimum reduc_salt across rooted layers becomes the plant salt stress factor strss.

salt_effect = 1 applies salinity stress after the other stresses are minimised; salt_effect = 2 (default) includes it in the joint minimum.

Salt uptake by plants

When salt_uptake_on = 1, daily salt removal by plants is read from a user file (salt_uptake.slt) and applied through salt_uptake.f90. Otherwise no plant uptake of salt occurs.

Trace constituent reactions and sorption

cs_rctn_hru.f90 applies first-order reactions to SeO4, SeO3, and boron. Coefficients come from cs_reactions.cs. cs_sorb_hru.f90 runs an equilibrium linear sorption between dissolved and solid-phase pools per layer, parameterised by partition coefficients in cs_data_module.f90. Constituent uptake follows the nitrate-style uptake routine cs_uptake.f90.

Salt and constituent inputs

- Atmospheric: wet deposition (salt_rain.f90, cs_rain.f90) and dry deposition (read from atmodep_salt, atmodep_cs), monthly or annual
- Fertiliser: salt-bearing fertilisers from salt_fert.f90, cs_fert.f90 (read into fert_salt, fert_cs)
- Road salt: salt_roadsalt.f90, applied seasonally per HRU
- Irrigation: salt and cs concentrations in irrigation water (salt_irrig.f90, cs_irrig.f90)
- Initial soil conditions: salt_hru_init.f90, cs_hru_init.f90

In-channel and aquifer

Channels route salt ions through ch_salt_module.f90 and ch_salt_output.f90; constituents through ch_cs_module.f90 and ch_cs_output.f90. Aquifer fluxes are handled in salt_aquifer.f90, salt_chem_aqu.f90, cs_aquifer.f90, cs_rctn_aqu.f90, cs_sorb_aqu.f90. Reservoirs and wetlands have parallel implementations (res_salt.f90, wet_salt.f90).

Switches and parameters

Salts

File	Field	Default	Effect
salt_module.f90	salt_uptake_on	0	1 = enable user-defined daily salt uptake
salt_data_module.f90	salt_tol_sim	0	1 = salinity stress on plant water uptake
salt_data_module.f90	salt_soil_type	0	1 = CaSO4 (gypsiferous), 2 = NaCl
salt_data_module.f90	salt_effect	0	1 = applied after other stresses, 2 = joint minimum
salt_data_module.f90	salt_tds_ec	input	TDS-to-EC conversion factor
salt_data_module.f90	Ksp11..Ksp51	see source	Soil mineral solubility products
salt_data_module.f90	Ksp12..Ksp52	same defaults	Aquifer mineral solubility products
salt_plants.slt	a, b per plant	input	Salt tolerance threshold and slope
salt_recall.slt and other recall files	input	Constituent loads on point sources	

Constituents (cs)

File	Field	Effect
cs_data_module.f90	partition coefficients	Linear sorption per constituent
cs_reactions.cs	first-order rate coefficients	Constituent reaction rates
cs_uptake.cs	uptake fractions per plant	Plant uptake of cs
cs_fert.cs	constituent loads in fertiliser	Inputs from fertiliser
cs_irrig.cs	constituent concentration in irrigation	Inputs from irrigation

Implementation

Salts

Soil and HRU: salt_balance.f90, salt_lch.f90, salt_chem_hru.f90, salt_chem_soil_single.f90, salt_uptake.f90, salt_rain.f90, salt_fert.f90, salt_fert_wet.f90, salt_irrig.f90, salt_roadsalt.f90, salt_hru_init.f90.

Inputs: salt_aqu_read.f90, salt_cha_read.f90, salt_fert_read.f90, salt_hru_read.f90, salt_irr_read.f90, salt_plant_read.f90, salt_roadsalt_read.f90, salt_uptake_read.f90, salt_urban_read.f90.

Modules: salt_module.f90, salt_data_module.f90.

Channel and water bodies: ch_salt_module.f90, ch_salt_output.f90, res_salt.f90, res_salt_module.f90, res_salt_output.f90, res_read_salt_cs.f90, res_read_saltdb.f90, wet_salt.f90, wet_salt_output.f90, wet_read_salt_cs.f90.

Aquifer: salt_aquifer.f90, salt_chem_aqu.f90, aqu_salt_output.f90.

Climate, recall, export coefficients, delivery ratios: cli_read_atmodep_salt.f90, recall_read_salt.f90, recall_salt.f90, dr_read_salt.f90, exco_read_salt.f90, header_salt.f90.

Output: hru_salt_output.f90, ru_salt_output.f90, output_ls_salt_module.f90.

Trace constituents (cs)

Soil and HRU: cs_balance.f90, cs_lch.f90, cs_rctn_hru.f90, cs_sorb_hru.f90, cs_uptake.f90, cs_rain.f90, cs_fert.f90, cs_fert_wet.f90, cs_irrig.f90, cs_hru_init.f90.

Inputs: cs_aqu_read.f90, cs_cha_read.f90, cs_fert_read.f90, cs_hru_read.f90, cs_irr_read.f90, cs_plant_read.f90, cs_reactions_read.f90, cs_uptake_read.f90, cs_urban_read.f90.

Modules: cs_module.f90, cs_data_module.f90.

Channel: ch_cs_module.f90, ch_cs_output.f90, cs_str_output.f90.

Aquifer: cs_aquifer.f90, cs_rctn_aqu.f90, cs_sorb_aqu.f90, aqu_cs_output.f90.

Reservoirs and wetlands: res_cs_module.f90, res_cs_output.f90, wet_cs_output.f90.

HRU and RU output: hru_cs_output.f90, ru_cs_output.f90.

Related pages

- Nutrient cycling
- Plant growth
- Aquifers
- Channels
- Recall and point sources